

Castalia iNQshop Design Document

Curriculum, sitemap, homepage wireframe, and concept plan for a one-car garage workshop classroom

Prepared as a concept brief for a toddler-plus homeschool makerspace.

Guiding idea: iNQshop should feel like a workshop for inquiry - safe for younger children, productive for older students, and rigorous enough to scale into real engineering, art, and research.

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1. Vision and design goals

iNQshop is the physical making complement to iNQspace. It should support craft, fabrication, observation, measurement, and publication across the whole educational arc - from early-childhood cardboard play through advanced fabrication and research-grade prototyping.

The room should not be designed as a conventional school classroom or a pure machine shop. It should read as a hybrid: part atelier, part laboratory, part tool library, and part family workshop. The core experience is that ideas become artifacts.

For this garage-based version, the design priority is to preserve the existing square center bench, keep the overhead garage door unobstructed, and create a safer step-up working surface for smaller children instead of movable stools.

Design goals

- Retain the center bench as the pedagogical heart of the room.
- Provide a supervised step-up platform for smaller children rather than loose stools.
- Place heavier, noisier, dusty, or fume-producing equipment on the perimeter walls.
- Keep the garage-door opening and circulation path clear.
- Support multiple modes at once: cardboard construction, painting, woodwork, digital fabrication, notebook work, and display.
- Make the room legible to children: tools, materials, exemplars, and project stages should be visible and named.

2. Curriculum architecture (K-PhD+)

The most powerful curriculum model is not a separate project list for every age. Instead, use the same project families at increasing depth: observe -> make -> measure -> model -> explain -> publish -> teach.

Strand	Seed / Apprentice	Builder / Studio	Collegium / Magisterium / PhD+
Form & structure	Towers, bridges, shelters, shape hunts	Trusses, joints, scale drawings, load tests	Parametric structures, material optimization, publishable studies
Motion & mechanism	Ramps, wheels, marble runs, automata	Linkages, gears, cams, motorized builds	Controls, robotics, mechanism synthesis
Light & image	Shadows, color, pinhole viewers	Optics, projection, stop-motion, scanning	Computational imaging, AR/VR interfaces
Matter & materials	Texture, paint, glue, clay, paper	Material comparisons, 3D print studies, laser settings	Characterization, sustainability, materials informatics
Nature & life	Habitats, nests, seeds, weather tools	Biomimicry, sensing, ecology builds	Bio-inspired engineering, field instruments
Civilization & artifact	Masks, maps, costumes, story	Historical reconstructions,	Research installations, public

	boxes	models, exhibit builds	humanities artifacts
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Core outputs for every level

- Artifact - the built thing
- Notebook - sketches, measurements, reflections
- Model - CAD, diagram, theory, or simulation
- Demonstration - show how it works
- Explanation - oral or written defense
- Publication - portfolio page, poster, paper, or guide
- Teaching - hand the knowledge downward to a younger student

3. One-car garage concept plan

This plan is a concept layout for a typical one-car garage footprint of roughly 12 ft x 20 ft. It is not a permit drawing and should be adapted to actual measurements, door swings, outlets, egress, and local code. The intent is to show zoning, circulation, and curriculum use.

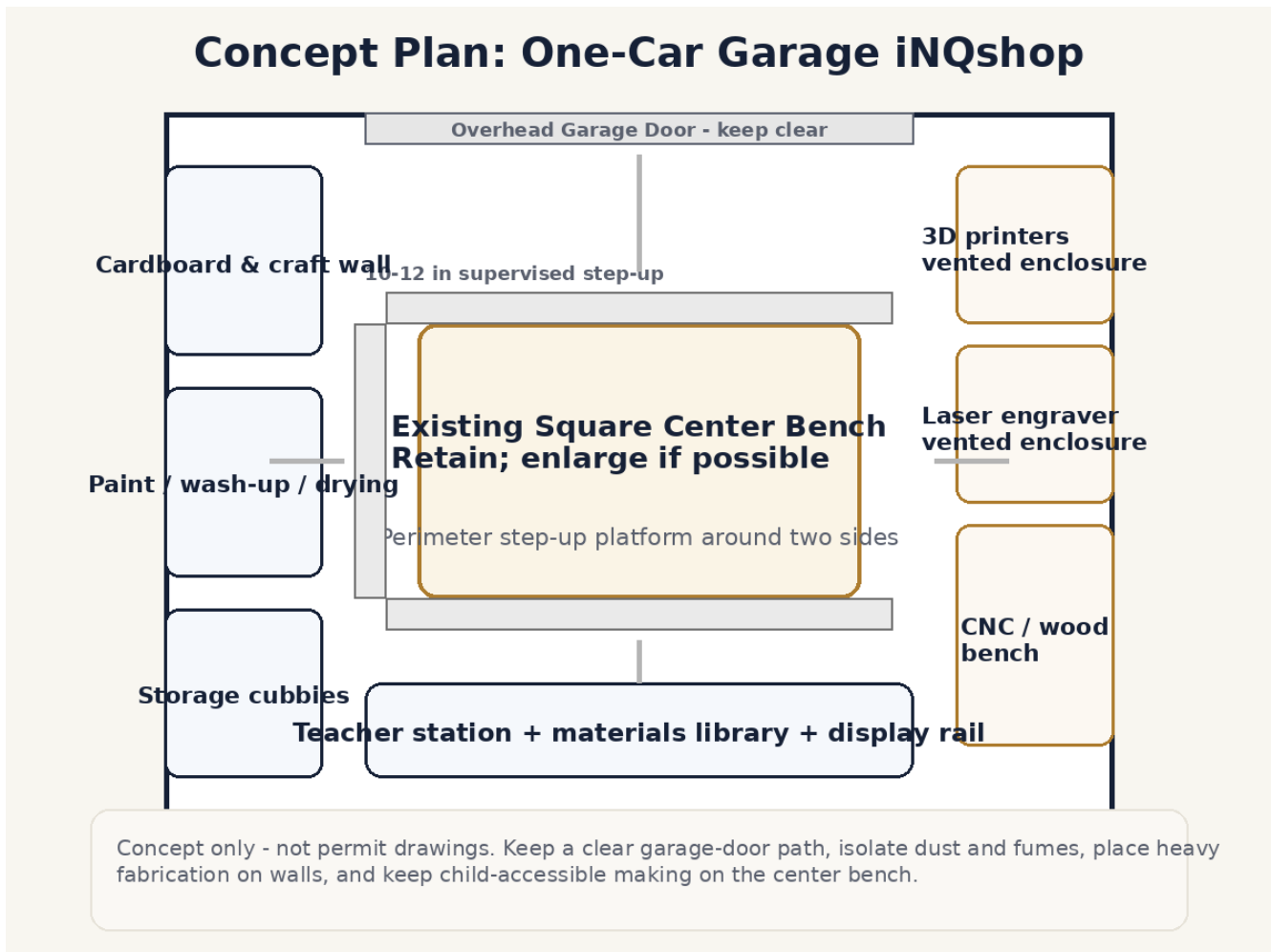


Figure 1. Concept plan for a one-car garage iNQshop.

Spatial logic

- The center bench remains the main shared work surface. It is where cardboard construction, drawing, notebook work, assembly, and supervised hand-tool work happen.
- A fixed or semi-fixed step-up surface (about 10-12 in high, with anti-slip finish and rounded edges) can wrap selected sides of the bench so smaller children can work at bench height without individual stools.
- The left wall should carry the softer, messier, child-forward zones: cardboard editing, adhesives, paint, drying racks, and open cubbies.
- The right wall should carry the fabrication edge: 3D printers in a vented enclosure, laser engraver in a vented enclosure, then CNC/wood bench with dust control.
- The back wall becomes the teaching and reference wall: teacher station, materials library, peg rail, project display, and current-work boards.
- The garage door opening must remain visually and physically clear. Do not park tall storage, carts, or enclosures in that path.

Suggested dimensional allowances

Element	Target	Notes
Center bench	Retain existing; enlarge if practical	Square plan preferred for group work
Step-up platform	10-12 in high, 12-18 in deep	Supervised use; anti-slip; rounded edges
Main circulation lane	At least 36 in clear	More if carts or rolling bins will pass
Laser enclosure	Dedicated wall bay	Vent directly outdoors if possible
3D print enclosure	Dedicated wall bay	Noise and filament management
CNC / wood bench	Most robust wall section	Dust collection and sacrificial top
Drying / display rail	Back or left wall	Keep wet work off main bench

4. Equipment, storage, safety, and utilities

Recommended room zones

Zone	Key contents
Cardboard Construction	corrugate stock, templates, child-safe cutters, metal rulers, hole punches, tabs/slots, hot-glue station for adult use only
Paint and Wet Media	washable paints, palettes, aprons, drying rack, wipe-clean mat, sink or utility wash tub if available
Wood and Hand Tools	saws appropriate to age, clamps, bench hooks, sanding zone, wood offcut bins
Digital Fabrication	3D printers, laser engraver, CNC router or desktop mill, tool calibration drawer, maintenance kit
Notebook and Design	graph paper, field notebooks, pencils, calipers, scales, examples, build instructions
Display and Archive	finished work shelf, current-project rail, labeled bins, material samples, exemplar board

Safety baseline

- Separate child-accessible tools from adult-only cutting, heat, and machine tools.
- Use vented enclosures for laser and preferably for 3D printing.
- Plan dust collection for CNC and sanding. Fine dust should not share air casually with paint or toddler activity.
- Keep a fire extinguisher, first-aid kit, eye protection, hearing protection, and clear shutdown procedures visible.
- Use labeled shadow-board storage so tools can be visually checked in and out.
- Treat the step-up surface as supervised infrastructure, not a play platform.

5. iNQspace sitemap

The front-end information architecture should feel more like a map of inquiry than a corporate LMS dashboard. The main navigational modes are Study, Make, Write, Explore, and Community.

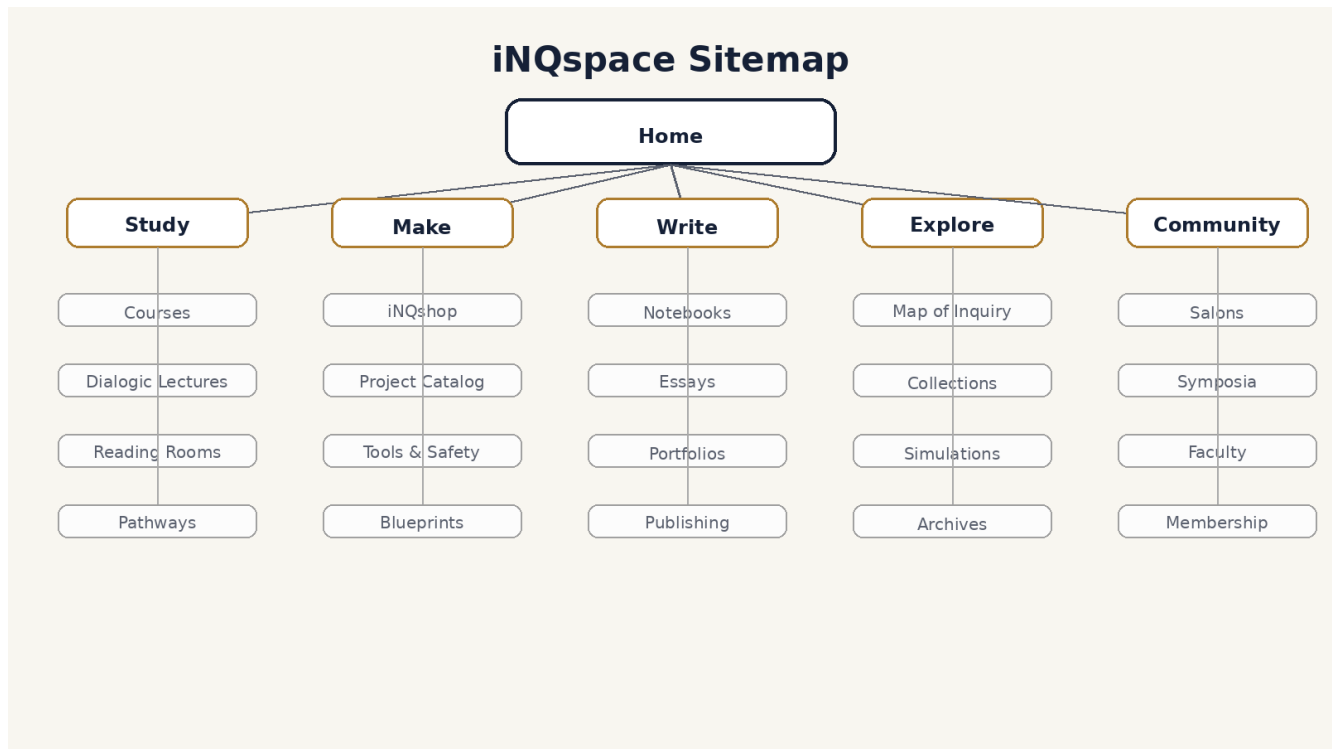


Figure 2. Proposed sitemap for the iNQspace ecosystem.

Recommended top-level navigation

- Home
- Study - courses, dialogic lectures, reading rooms, pathways
- Make - iNQshop, project catalog, tools and safety, garage plans and blueprints
- Write - notebooks, essays, portfolios, publishing
- Explore - map of inquiry, collections, simulations, archives
- Community - salons, symposia, faculty, membership

6. iNQspace homepage wireframe

The homepage should greet the visitor with a strong civilizational tone: not software first, but inquiry first. The interface can still be highly functional, but the visual grammar should evoke library, observatory, and workshop.

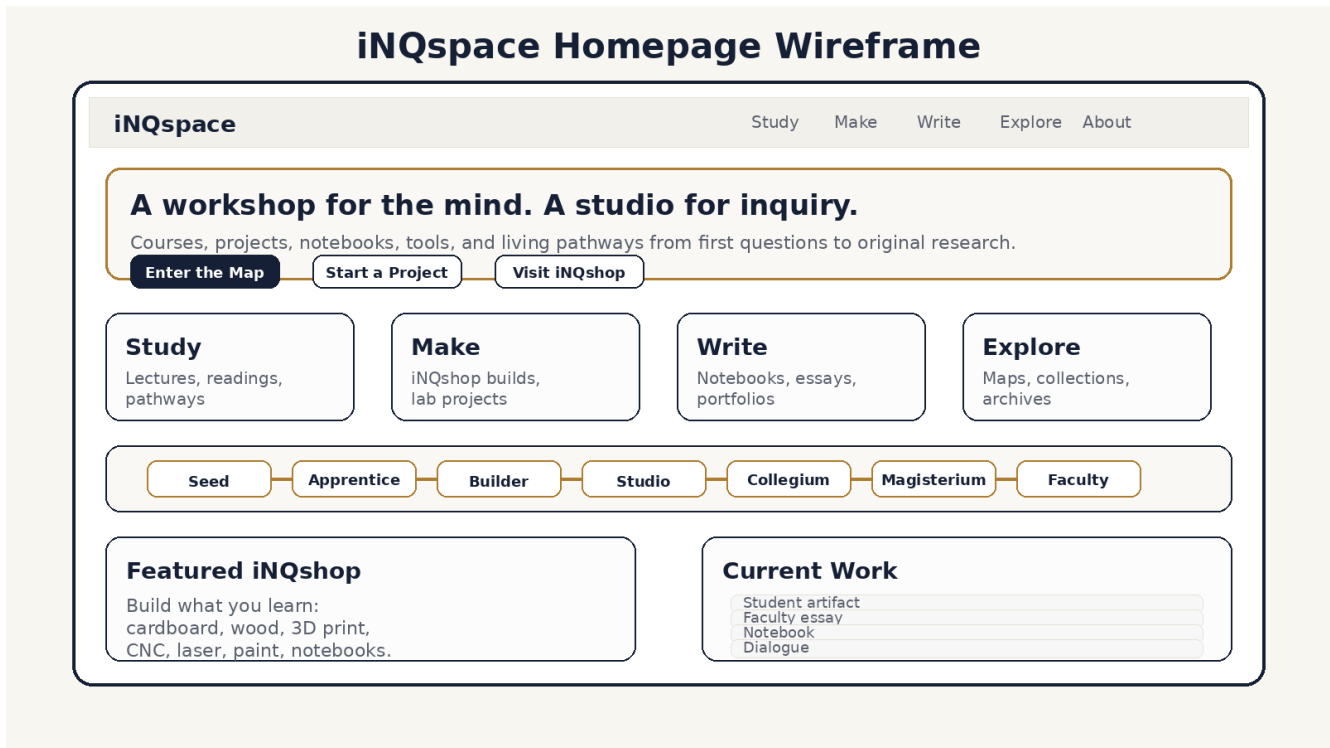


Figure 3. Homepage wireframe for iNQspace.

Homepage blocks

- Hero: A workshop for the mind. A studio for inquiry.
- Mode tiles: Study, Make, Write, Explore.
- Pathway ladder: Seed, Apprentice, Builder, Studio, Collegium, Magisterium, Faculty.
- Featured iNQshop module and current work area.
- Ecosystem footer linking Inquirer, Atlas, Symposia, Salons, and courses.

7. Blueprint packages and plan sets to offer

A strong Castalia offering is not only the room itself, but a family of reproducible plans. These should be sold or published as concept sets rather than represented as stamped construction documents.

Package	Audience	Contents	Use case
Starter Garage	Homeschool families	Floor plan, storage elevations, tool list, safety checklist, curriculum starter pack	Convert one bay quickly and affordably
Toddler + Builder	Families with mixed ages	Bench/step-up details, child zones, washable materials, low-risk tool progression	Shared family workshop
Fabrication Garage	Older learners / micro-school	Laser/CNC/3D print enclosure plan, power/vent guidance, advanced tool list	Small but serious makerspace
Mobile Pop-up	Co-op / temporary use	Fold-flat carts, portable surfaces, display boards, transport list	Deployable workshop
Studio + Shop Hybrid	Art + fabrication focus	Paint wall, print wall, book arts zone, digital fabrication edge	Balanced atelier / makerspace

Recommended drawing set for each package

- 1. Cover sheet with concept and audience
- 2. Measured plan with furniture and equipment layout
- 3. Wall elevations for storage, pegboards, rails, and cubbies
- 4. Bench and step-up surface details
- 5. Utilities and ventilation concept sheet
- 6. Safety and supervision sheet
- 7. Sample curriculum sheet linking zones to projects
- 8. Purchasing/specification sheet with optional good-better-best equipment tiers

8. Phased implementation roadmap

Phase 1 - Clear and zone

Paint, lighting, wall storage, center bench retention, cardboard and notebook capability

Phase 2 - Child-ready making

Step-up surface, aprons, drying racks, labeled bins, safe hand-tool set

Phase 3 - Fabrication edge

3D printers, laser enclosure, dust-aware wood/CNC bench, maintenance storage

Phase 4 - Curriculum publishing

Project ladder, blueprint product line, iNQspace pages, exemplar builds

Closing note

The real strength of this design is that the room, the curriculum, and the web front-end all say the same thing: learning should result in artifacts, evidence, and living competence. The garage becomes not just a room, but a visible node in the Castalia ecology - iNQspace online, iNQshop in the world.